

# Sourav Danyal

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## SUMMARY

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Motivated Data Science and AI enthusiast with hands-on experience in building ML models, RAG applications, and API services. Passionate about applying data-driven solutions to real-world problems.

## EDUCATION

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| <b>Central University of Himachal Pradesh</b><br><i>Master of Computer Applications (MCA)</i>           | 2024 – 2026 |
| <b>Government Degree College Shahpur (HPU)</b><br><i>B.Sc in Mathematics, Computer Science, Physics</i> | 2021 – 2024 |

## EXAMS QUALIFIED

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- Qualified **UGC NET (Computer Science)**, June 2025 – Eligible for PhD admission.

## PROJECTS

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- **Multi-Agent AI Research Assistant** [Live Demo](#) | [Github](#)  
Built a full-stack AI research assistant using FastAPI, React, LangGraph, LangChain, Groq LLaMA 3.1, Tavily Search, ChromaDB, and Hugging Face embeddings. Designed a multi-agent workflow with researcher, summarizer, writer, and fact-checker agents to generate sourced, verified research reports. Implemented real-time streaming with Server-Sent Events, document upload support, citation handling, and markdown report generation. Containerized the application with Docker and configured deployment-ready frontend/backend architecture.
- **DocRAG Studio: AI Document Analyzer** [Live Demo](#) | [GitHub](#)  
Built a production-ready full-stack RAG application using React, FastAPI, LangChain, ChromaDB (vector database), and Groq LLaMA 3.1 8B with Hugging Face embeddings for semantic search. Supports context-aware question answering across 5 document formats (PDF, CSV, Excel, TXT, DOCX) with per-file context isolation and source-cited responses. Containerized with Docker and exposed RESTful APIs for scalable, production-ready document ingestion and querying.
- **Movie Recommendation System** [Live Demo](#) | [GitHub](#)  
Developed a content-based movie recommendation engine using cosine similarity on the TMDB dataset of 5,000+ films, built with Python, Pandas, and Scikit-learn. Deployed an interactive Streamlit web app with real-time poster and trailer retrieval via TMDB API, cached API calls for performance, and secure environment-based API key management.

## HACKATHONS

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- **Meta PyTorch OpenEnv Hackathon** – Finalist. Out of 31,000+ team registrations, our team was among the Top 800 teams advancing to the finale organized at Scaler School of Technology (Bangalore).
- **Tryst (IIT Delhi)** – Secured Second Place in a Mathematical Model competition related to Machine Learning.
- **Chaitanya (Himachal Pradesh Technical University)** – Secured Second Place in a Datathon (Machine Learning).

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C/C++, Java, SQL
- **Data Science & Generative AI:** Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit
- **Frameworks:** TensorFlow, PyTorch, LangChain, FastAPI, Flask
- **Databases:** MySQL, MongoDB, ChromaDB
- **Core Concepts:** OOP, DSA, Operating Systems, DBMS, Rest APIs
- **Deployment & Tools:** Docker, Hugging Face Spaces, Git, GitHub, VS Code, Jupyter Notebook, Vercel